

Electronics & Computer Science

COMP 1201 Algorithmics

Tutorial 10

1. Consider the optimisation problem below. Find the solution using the simplex algorithm and graphically. Does the problem have more than one solution?

$$\begin{aligned} &\text{maximize } 6x_1 + 3x_2 \\ &\text{subject to } 2x_1 + x_2 \leq 20 \\ &\quad \quad \quad x_1 - x_2 \leq 4 \\ &\quad \quad \quad 3x_1 + 2x_2 \leq 35 \\ &\quad \quad \quad x_1, x_2 \geq 0 \end{aligned}$$

2. Solve the following LP using the simplex algorithm.

$$\begin{aligned} &\text{maximize } 5x_1 + 3x_2 \\ &\text{subject to } 2x_1 + x_2 \leq 20 \\ &\quad \quad \quad x_1 - x_2 \leq -16 \\ &\quad \quad \quad 3x_1 + 2x_2 \leq 35 \\ &\quad \quad \quad x_1, x_2 \geq 0 \end{aligned}$$

3. Convert the optimisation problem below into standard form

$$\begin{aligned} &\text{minimise } 2x_1 + 7x_2 + x_3 \\ &\text{subject to } x_1 - x_3 = 7 \\ &\quad \quad \quad 3x_1 + x_2 \geq 24 \\ &\quad \quad \quad x_2 \geq 0 \\ &\quad \quad \quad x_3 \leq 0 \end{aligned}$$

4. Show that the optimisation problem below is infeasible

$$\begin{aligned} &\text{maximise } 3x_1 - 2x_2 \\ &\text{subject to } \\ &\quad \quad \quad x_1 + x_2 \leq 2 \\ &\quad \quad \quad -2x_1 - 2x_2 \leq -10 \\ &\quad \quad \quad x_1, x_2 \geq 0 \end{aligned}$$

5. Consider the following LP

$$\begin{array}{ll}\text{maximise} & -x_1 - x_2 \\ \text{subject to} & -2x_1 - x_2 \leq 4 \\ & -2x_1 + 4x_2 \leq -8 \\ & -x_1 + 3x_2 \leq -7 \\ & x_1, x_2 \geq 0\end{array}$$

- (a) What is the dual of the above LP problem?
- (b) Write both the primal and dual in slack form.
- (c) Solve the dual using the simplex algorithm.
- (d) Complementary slackness. Let $x_1, \dots, x_n$ and $u_1, \dots, u_m$ be the variables/slack variables for the primal and $y_1, \dots, y_m$ and $w_1, \dots, w_n$ the variables/slack variables for the dual. Complementary slackness states that if both the primal and dual are feasible then $x_i w_i = 0$ and $y_j u_j = 0$ for all $i = 1 \dots n, j = 1 \dots m$.
Use the solution for the dual and complementary slackness to find the solution for the primal.